

OpenStax Calculus Volume 3

Section 2.2 Exercises: Vectors in Three Dimensions

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1. Consider a rectangular box with one of the vertices at the origin, as shown in the following figure. If point $A(2, 3, 5)$ is the opposite vertex to the origin, then find
[Figure: This figure is the first octant of the 3-dimensional coordinate system. It has a point labeled “A(2, 3, 5)” drawn.]
 - (a) the coordinates of the other six vertices of the box and
 - (b) the length of the diagonal of the box determined by the vertices O and A .
2. Find the coordinates of point P and determine its distance to the origin.
[Figure: This figure is the first octant of the 3-dimensional coordinate system. It has a point drawn at $(2, 1, 1)$. The point is labeled “P.”]

For the following exercises, describe and graph the set of points that satisfies the given equation.

3. $(y - 5)(z - 6) = 0$
4. $(z - 2)(z - 5) = 0$
5. $(y - 1)^2 + (z - 1)^2 = 1$
6. $(x - 2)^2 + (z - 5)^2 = 4$
7. Write the equation of the plane passing through point $(1, 1, 1)$ that is parallel to the xy -plane.
8. Write the equation of the plane passing through point $(1, -3, 2)$ that is parallel to the xz -plane.
9. Find an equation of the plane passing through points $(1, -3, -2)$, $(0, 3, -2)$, and $(1, 0, -2)$.
10. Find an equation of the plane passing through points $(1, 9, 2)$, $(1, 3, 6)$, and $(1, -7, 8)$.

For the following exercises, find an equation of the sphere in standard form that satisfies the given conditions.

11. Center $C(-1, 7, 4)$ and radius 4
12. Center $C(-4, 7, 2)$ and radius 6
13. Diameter PQ , where $P(-1, 5, 7)$ and $Q(-5, 2, 9)$
14. Diameter PQ , where $P(-16, -3, 9)$ and $Q(-2, 3, 5)$

For the following exercises, find the center and radius of the sphere with an equation in general form that is given.

15. $x^2 + y^2 + z^2 - 4z + 3 = 0$

16. $x^2 + y^2 + z^2 - 6x + 8y - 10z + 25 = 0$

For the following exercises, express vector $\vec{PQ}$ with the initial point at P and the terminal point at Q

(a) in component form and

(b) by using standard unit vectors.

17. $P(3, 0, 2)$ and $Q(-1, -1, 4)$

18. $P(0, 10, 5)$ and $Q(1, 1, -3)$

19. $P(-2, 5, -8)$ and $M(1, -7, 4)$, where M is the midpoint of the line segment PQ

20. $Q(0, 7, -6)$ and $M(-1, 3, 2)$, where M is the midpoint of the line segment PQ

21. Find terminal point Q of vector $\vec{PQ} = \langle 7, -1, 3 \rangle$ with the initial point at $P(-2, 3, 5)$.

22. Find initial point P of vector $\vec{PQ} = \langle -9, 1, 2 \rangle$ with the terminal point at $Q(10, 0, -1)$.

For the following exercises, use the given vectors $\mathbf{a}$ and $\mathbf{b}$ to find and express the vectors $\mathbf{a} + \mathbf{b}$, $4\mathbf{a}$, and $-5\mathbf{a} + 3\mathbf{b}$ in component form.

23. $\mathbf{a} = \langle -1, -2, 4 \rangle$, $\mathbf{b} = \langle -5, 6, -7 \rangle$

24. $\mathbf{a} = \langle 3, -2, 4 \rangle$, $\mathbf{b} = \langle -5, 6, -9 \rangle$

25. $\mathbf{a} = -\mathbf{k}$, $\mathbf{b} = -\mathbf{i}$

26. $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$

For the following exercises, vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Find the magnitudes of vectors $\mathbf{u} - \mathbf{v}$ and $-2\mathbf{u}$.

27. $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{v} = -\mathbf{i} + 5\mathbf{j} - \mathbf{k}$

28. $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} - \mathbf{k}$

29. $\mathbf{u} = \langle 2 \cos t, -2 \sin t, 3 \rangle$, $\mathbf{v} = \langle 0, 0, 3 \rangle$, where t is a real number.

30. $\mathbf{u} = \langle 0, 1, \sinh t \rangle$, $\mathbf{v} = \langle 1, 1, 0 \rangle$, where t is a real number.

For the following exercises, find the unit vector in the direction of the given vector $\mathbf{a}$ and express it using standard unit vectors.

31. $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$

32. $\mathbf{a} = \langle 4, -3, 6 \rangle$

33. $\mathbf{a} = \vec{PQ}$, where $P(-2, 3, 1)$ and $Q(0, -4, 4)$

34. $\mathbf{a} = \vec{OP}$, where $P(-1, -1, 1)$

35. $\mathbf{a} = \mathbf{u} - \mathbf{v} + \mathbf{w}$, where $\mathbf{u} = \mathbf{i} - \mathbf{j} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, and $\mathbf{w} = -\mathbf{i} + \mathbf{j} + 3\mathbf{k}$
36. $\mathbf{a} = 2\mathbf{u} + \mathbf{v} - \mathbf{w}$, where $\mathbf{u} = \mathbf{i} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{j}$, and $\mathbf{w} = \mathbf{i} - \mathbf{j}$
37. Determine whether $\vec{AB}$ and $\vec{PQ}$ are equivalent vectors, where $A(1, 1, 1)$, $B(3, 3, 3)$, $P(1, 4, 5)$, and $Q(3, 6, 7)$.
38. Determine whether the vectors $\vec{AB}$ and $\vec{PQ}$ are equivalent, where $A(1, 4, 1)$, $B(-2, 2, 0)$, $P(2, 5, 7)$, and $Q(-3, 2, 1)$.

For the following exercises, find vector $\mathbf{u}$ with a magnitude that is given and satisfies the given conditions.

39. $\mathbf{v} = \langle 7, -1, 3 \rangle$, $\|\mathbf{u}\| = 10$, $\mathbf{u}$ and $\mathbf{v}$ have the same direction
40. $\mathbf{v} = \langle 2, 4, 1 \rangle$, $\|\mathbf{u}\| = 15$, $\mathbf{u}$ and $\mathbf{v}$ have the same direction
41. $\mathbf{v} = \langle 2 \sin t, 2 \cos t, 1 \rangle$, $\|\mathbf{u}\| = 2$, $\mathbf{u}$ and $\mathbf{v}$ have opposite directions for any t , where t is a real number
42. $\mathbf{v} = \langle 3 \sinh t, 0, 3 \rangle$, $\|\mathbf{u}\| = 5$, $\mathbf{u}$ and $\mathbf{v}$ have opposite directions for any t , where t is a real number
43. Determine a vector of magnitude 5 in the direction of vector $\vec{AB}$, where $A(2, 1, 5)$ and $B(3, 4, -7)$.
44. Find a vector of magnitude 2 that points in the opposite direction than vector $\vec{AB}$, where $A(-1, -1, 1)$ and $B(0, 1, 1)$. Express the answer in component form.
45. Consider the points $A(2, \alpha, 0)$, $B(0, 1, \beta)$, and $C(1, 1, \beta)$, where α and β are negative real numbers. Find α and β such that $\|\vec{OA} - \vec{OB} + \vec{OC}\| = \|\vec{OB}\| = 4$.
46. Consider points $A(\alpha, 0, 0)$, $B(0, \beta, 0)$, and $C(\alpha, \beta, \beta)$, where α and β are positive real numbers. Find α and β such that $\|\vec{OA} + \vec{OB}\| = \sqrt{2}$ and $\|\vec{OC}\| = \sqrt{3}$.
47. Let $P(x, y, z)$ be a point situated at an equal distance from points $A(1, -1, 0)$ and $B(-1, 2, 1)$. Show that point P lies on the plane of equation $-2x + 3y + z = 2$.
48. Let $P(x, y, z)$ be a point situated at an equal distance from the origin and point $A(4, 1, 2)$. Show that the coordinates of point P satisfy the equation $8x + 2y + 4z = 21$.
49. The points A , B , and C are collinear (in this order) if the relation $\|\vec{AB}\| + \|\vec{BC}\| = \|\vec{AC}\|$ is satisfied. Show that $A(5, 3, -1)$, $B(-5, -3, 1)$, and $C(-15, -9, 3)$ are collinear points.
50. Show that points $A(1, 0, 1)$, $B(0, 1, 1)$, and $C(1, 1, 1)$ are not collinear.
51. [T] A force $\mathbf{F}$ of 50 N acts on a particle in the direction of the vector $\vec{OP}$, where $P(3, 4, 0)$.
 - (a) Express the force as a vector in component form.
 - (b) Find the angle between force $\mathbf{F}$ and the positive direction of the x -axis. Express the answer in degrees rounded to the nearest integer.
52. [T] A force $\mathbf{F}$ of 40 N acts on a box in the direction of the vector $\vec{OP}$, where $P(1, 0, 2)$.
 - (a) Express the force as a vector by using standard unit vectors.

- (b) Find the angle between force F and the positive direction of the x -axis.
53. If F is a force that moves an object from point $P_1(x_1, y_1, z_1)$ to another point $P_2(x_2, y_2, z_2)$, then the displacement vector is defined as $D = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$. A metal container is lifted 10 m vertically by a constant force F . Express the displacement vector D by using standard unit vectors.
54. A box is pulled 4 yd horizontally in the x -direction by a constant force F . Find the displacement vector in component form.
55. The sum of the forces acting on an object is called the resultant or net force. An object is said to be in static equilibrium if the resultant force of the forces that act on it is zero. Let $F_1 = \langle 10, 6, 3 \rangle$, $F_2 = \langle 0, 4, 9 \rangle$, and $F_3 = \langle 10, -3, -9 \rangle$ be three forces acting on a box. Find the force F_4 acting on the box such that the box is in static equilibrium. Express the answer in component form.
56. [T] Let $F_k = \langle 1, k, k^2 \rangle$, $k = 1, \dots, n$ be n forces acting on a particle, with $n \geq 2$.
- (a) Find the net force $F = \sum_{k=1}^n F_k$. Express the answer using standard unit vectors.
- (b) Use a computer algebra system (CAS) to find n such that $\|F\| < 100$.
57. The force of gravity F acting on an object is given by $F = mg$, where m is the mass of the object (expressed in kilograms) and g is acceleration resulting from gravity, with $\|g\| = 9.8 \text{ N/kg}$. A 2-kg disco ball hangs by a chain from the ceiling of a room.
- (a) Find the force of gravity F acting on the disco ball and find its magnitude.
- (b) Find the force of tension T in the chain and its magnitude.
Express the answers using standard unit vectors.
58. A 5-kg pendant chandelier is designed such that the alabaster bowl is held by four chains of equal length, as shown in the following figure.
[Figure: This figure shows a light fixture hung from a ceiling, supported by 4 chains from the same point on the ceiling to four points spread evenly around the light fixture.]
- (a) Find the magnitude of the force of gravity acting on the chandelier.
- (b) Find the magnitudes of the forces of tension for each of the four chains (assume chains are essentially vertical).
59. [T] A 30-kg block of cement is suspended by three cables of equal length that are anchored at points $P(-2, 0, 0)$, $Q(1, \sqrt{3}, 0)$, and $R(1, -\sqrt{3}, 0)$. The load is located at $S(0, 0, -2\sqrt{3})$, as shown in the following figure. Let F_1 , F_2 , and F_3 be the forces of tension resulting from the load in cables RS , QS , and PS , respectively.
[Figure: This figure is the 3-dimensional coordinate system. It has 4 points drawn. The first point is labeled "P(-2, 0, 0)." The second point is labeled "R(1, -squeroot of 3, 0)." The third point is labeled "S(0, 0, -2squeroot of 3)." The fourth point is labeled "Q(1, squareroot of 3, 0)." There are line segments from P to S, from R to S, and from Q to S. At point S there is a box labeled "30 kg."]
- (a) Find the gravitational force F acting on the block of cement that counterbalances the sum $F_1 + F_2 + F_3$ of the forces of tension in the cables.
- (b) Find forces F_1 , F_2 , and F_3 . Express the answer in component form.

60. Two soccer players are practicing for an upcoming game. One of them runs 10 m from point A to point B. She then turns left at 90° and runs 10 m until she reaches point C. Then she kicks the ball with a speed of 10 m/sec at an upward angle of 45° to her teammate, who is located at point A. Write the velocity of the ball in component form.
- [Figure: This figure is the image of two soccer players. The first soccer player is at point A. The second player is at point C. There is a line segment from A to C. There is a vector from player C upwards labeled "v." There is a vector from player A to the bottom of the image. The point at the bottom is labeled "B." This vector is labeled "10m." There is a vector from C to B labeled "10m."]*
61. Let $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ be the position vector of a particle at the time $t \in [0, T]$, where x , y , and z are smooth functions on $[0, T]$. The instantaneous velocity of the particle at time t is defined by vector $\mathbf{v}(t) = \langle x'(t), y'(t), z'(t) \rangle$, with components that are the derivatives with respect to t , of the functions x , y , and z , respectively. The magnitude $\|\mathbf{v}(t)\|$ of the instantaneous velocity vector is called the speed of the particle at time t . Vector $\mathbf{a}(t) = \langle x''(t), y''(t), z''(t) \rangle$, with components that are the second derivatives with respect to t , of the functions x , y , and z , respectively, gives the acceleration of the particle at time t . Consider $\mathbf{r}(t) = \langle \cos t, \sin t, 2t \rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\mathbf{r}$ are expressed in centimeters and time is expressed in seconds.
- Find the instantaneous velocity, speed, and acceleration of the particle after the first second. Round your answer to two decimal places.
 - Use a CAS to visualize the path of the particle—that is, the set of all points of coordinates $(\cos t, \sin t, 2t)$, where $t \in [0, 30]$.
62. [T] Let $\mathbf{r}(t) = \langle t, 2t^2, 4t^2 \rangle$ be the position vector of a particle at time t (in seconds), where $t \in [0, 10]$ (here the components of $\mathbf{r}$ are expressed in centimeters).
- Find the instantaneous velocity, speed, and acceleration of the particle after the first two seconds. Round your answer to two decimal places.
 - Use a CAS to visualize the path of the particle defined by the points $(t, 2t^2, 4t^2)$, where $t \in [0, 10]$.